

ALEXANDRE DELLAL^{1,2} AND DEL P. WONG³

ABSTRACT

between RSA and RCOD, i.e., the RSA/RCOD index. In the study of Wong et al. (34), they introduced the concept of RSA/RCOD index, a relative, norm-referenced index that help providing individualized suggestions for improvement in RSA and RCOD. It was found that physical active individuals have significantly longer fastest time (FT), average time (AT), and total time (TT) in both RSA and RCOD compared with soccer players. It was further reported that the RSA/RCOD index of soccer players is 0.59 compared with 0.53 for physical active individuals (34). However, to our knowledge, there is no study comparing the RSA/RCOD index of soccer players from different age groups.

Therefore, the purpose of the present study was to compare the RSA/RCOD index among soccer players with different age groups from an elite team. It is hypothesized that compared with RSA, RCOD is a more complex movement that is not yet fully developed at early ages, and thus different RSA/RCOD index values may be used to guide the training priority at different age groups.

METHODS

Experimental Approach to the Problem

Elite soccer players from the same club and from different age groups (U15, U17, U19, and Pro2) were tested in both a RSA (10 × 20 m with active recovery) and a RCOD test (10 × 20 m with active recovery) during the competitive phase (October). The distance, number of interval, and recovery duration were all equated among the RSA and RCOD tests. The RSA and RCOD tests were performed on the same synthetic turf of last generation in similar environmental conditions (temperature: 12.1° C vs. 13.2° C and relative humidity: 67.9% vs. 70.0%) at the same time of the day to avoid any effect of circadian rhythms (7) and with 7 days

separating the 2 tests. All players have the same nutritional intakes and sleep time. A standardized 20-minute warm-up was performed before each test including a solicitation of the hamstring, gluteus, quadriceps, gastrocnemius muscle, active-dynamic stretching, and different exercises of short sprints distance or directional changes. Consistent verbal encouragement was provided by the coaches for all the RSA and RCOD tests to ensure the maintenance of a consistently high work rate.

Subjects

Forty-nine elite soccer players from the French Division One football club (93% of them began the soccer in club at 7.1 years old) and from different age categories [second team (national competition: CFA-CFA2) $n = 8$, U19: $n = 18$, U17: $n = 13$, and U15: $n = 10$] participated in this study. Players' anthropometrics values and playing positions are presented in Table 1. They participated between 4 and 7 training sessions plus 1 official games each weeks. Goalkeepers were excluded from the study because they did not participate in the same physical training program as the remainder of the squad. The study was conducted according to the ethical standards in sport and exercise science research (16) and the protocol was fully approved by the Human Research Ethics Committee before the commencement of the assessments. All subjects received a detailed explanation about the aims of the study. The players and their parents gave their informed consent as part of their sport requirements, which is consistent with our institution's policies for use of human subject research.

RSA and RCOD Tests

The RSA and RCOD tests were performed with 10 repetitions of 20 m with 25 seconds of active recovery between each repetition (34). The RSA test was applied in straight-line sprints, whereas RCOD test consisted of four 100° COD at every 4 m (Figure 1). The active recovery consisted of jogged slowly back to the starting line and waited for the next sprint providing by an auditory signal. Performance was measured using an infrared timing system (Polifemo; Microgate, Bolzano-Bozen, Italy) placed at the starting line and the finishing line in a height corresponding to the player's pelvis.

The AT, FT, and TT were recorded during all the RSA and RCOD tests according to previous studies (21,34). The percentage of decrement score (%Dec) was then calculated using the formula proposed by Glaister (13), which has been

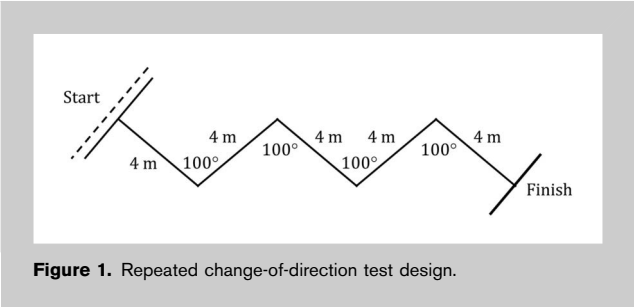
TABLE 1. Means (SD) of anthropometrics measures and the repartition of the playing position in the 4 groups.

Categories	U15	U17	U19	Pro2
% Body fat	10.67*† (1.54)	8.84 (0.85)	9.93 (1.65)†	7.87 (0.71)
Body mass (kg)	53.40 (9.87)*†‡	67.25 (4.70)	68.96 (6.38)	72.38 (4.27)
Height (m)	1.67 (0.08)*†‡	1.78 (0.06)	1.78 (0.07)	1.80 (0.04)
BMI (kg·m ⁻²)	18.96 (2.09)*†‡	21.23 (1.44)	21.70 (1.60)	22.34 (1.48)
Central defenders (%)	10.0	15.4	22.2	12.5
Full Backs (%)	50.0	7.7	16.7	12.5
Axial midfielders (%)	30.0	23.1	16.7	25.0
External midfielders (%)	10.0	30.8	22.2	25.0
Offensive axial midfielders (%)	0.0	0.0	11.1	12.5
Forwards (%)	0.0	23.1	11.1	12.5

*Significant different from U17 at $p < 0.05$.

†Significant different from Pro2 at $p < 0.05$.

‡Significant different from U19 at $p < 0.05$.



demonstrated as the most valid and reliable method of quantifying fatigue in multiple sprints test (13). It has been reported that the RSA and RCOD are reliable measures (Coefficient of Variation <10% and intra-class correlation coefficient >0.80) (34). Then, the RSA/RCOD index proposed by Wong et al. (34) was calculated for the different parameters analyzed (AT, FT, TT, and %Dec), i.e., the index-AT, index-FT, index-TT, and index-%Dec.

Statistical Analyses

An one-way analysis of variance (ANOVA) was used to examine the differences in percentage of body fat, body mass, height, body mass index (BMI), and the RSA/RCOD index between the 4 age groups. When a significant difference was found in the above analysis, pairwise comparisons were made using Bonferroni's adjustment to control the type-1 error rate. The significant level was set as $p < 0.05$.

RESULTS

One-way ANOVA results (Table 1) showed significant difference between groups in percentage body fat ($F = 8.21$, $p < 0.01$), body mass ($F = 15.99$, $p < 0.01$), height ($F = 7.78$, $p < 0.01$), and BMI ($F = 7.91$, $p < 0.01$). Compared with the other groups, the U15 players had significant high percentage of body fat ($p < 0.05$), lower body mass ($p < 0.05$), shorter body height ($p < 0.05$), and lower BMI ($p < 0.05$).

Two-way ANOVA results showed significant group effects on RSA parameters ($F = 36.34$, $p < 0.01$, Table 2), RCOD parameters ($F = 22.73$, $p < 0.01$, Table 2), and RSA/RCOD index parameters ($F = 22.27$, $p < 0.01$, Table 3). Compared with other groups, U15 has significant ($p < 0.05$) higher values in RSA-FT, RSA-AT, RSA-TT, RCOD-FT, RCOD-AT, RCOD-TT, index-FT, index-AT, and index-TT. Likewise, U17 has significant ($p < 0.05$) higher values in RSA-FT, RSA-AT, RSA-TT, RCOD-FT, RCOD-AT, RCOD-TT, index-FT, index-AT, and index-TT compared with U19 and Pro2. No difference concerning these parameters analyzed was found between U19 and Pro2. However, the trend of performance time and RSA/RCOD index was always similar (i.e., $U15 > U17 > U19 > Pro2$, $Pro2 > U19 > U17 > U15$) showing an age dependant for RSA and RCOD performance.

The percentage of decrement score in RSA of U15 and U17 was significantly ($p < 0.05$, Table 2) higher than U19

TABLE 2. Means (SD) of all measures in the 4 groups.

Group	RSA-FT (s)	RSA-AT (s)	RSA-TT (s)	RSA-%Dec (%)	RCOD-FT (s)	RCOD-AT (s)	RCOD-TT (s)	RCOD-%Dec (%)
U15	3.29 (0.06)*†‡	3.34 (0.06)*†‡	20.05 (0.38)*†‡	1.72 (0.54)†‡	5.45 (0.04)*†‡	5.54 (0.05)*†‡	33.21 (0.28)*†‡	1.57 (0.46)†
U17	3.14 (0.08)†‡	3.19 (0.09)†‡	19.13 (0.52)†‡	1.62 (0.47)†‡	5.39 (0.03)†‡	5.46 (0.04)†‡	32.76 (0.24)†‡	1.39 (0.56)
U19	3.01 (0.10)	3.04 (0.10)	18.23 (0.60)	1.00 (0.42)	5.34 (0.04)	5.39 (0.04)	32.35 (0.26)	0.95 (0.36)
Pro2	2.93 (0.07)	2.96 (0.08)	17.75 (0.46)	0.94 (0.37)	5.31 (0.05)	5.37 (0.07)	32.22 (0.42)	1.24 (0.45)

*Significant different from U17 at $p < 0.05$.
†Significant different from U19 at $p < 0.05$.
‡Significant different from Pro2 at $p < 0.05$.
RSA = repeated sprint ability; RCOD = repeated change-of-direction; FT = the fastest time; AT = average time; TT = total time; %Dec = percentage of the decrement score.

TABLE 3. Comparison of RSA/RCOD index among the 4 groups.

Groups	RSA/RCOD index			
	FT	At	TT	%Dec
U15	0.60 (0.01)*†‡	0.60 (0.01)*†‡	0.60 (0.01)*†‡	1.19 (0.48)
U17	0.58 (0.01)†‡	0.58 (0.01)†‡	0.58 (0.01)†‡	1.37 (0.65)
U19	0.56 (0.02)	0.56 (0.02)	0.56 (0.02)	1.19 (0.66)
Pro	0.55 (0.01)	0.55 (0.02)	0.55 (0.02)	0.90 (0.64)

*Significant different from U17 at $p < 0.05$.

†Significant different from U19 at $p < 0.05$.

‡Significant different from Pro at $p < 0.05$.

RSA = repeated sprint ability; RCOD = repeated change-of-direction; FT = the fastest time; AT = average time; TT = total time; %Dec = percentage decrement.

and Pro2. In addition, the percentage of decrement score in RCOD of U15 was significantly ($p < 0.05$) higher than U19.

DISCUSSION

The aim of the present study was to compare the RSA/RCOD index among elite soccer players with different age groups (U15, U17, U19, and Pro2). The main findings showed that the performance in RSA and RCOD are affected by the age of the players, especially with the larger difference between U15 and U19/Pro2 categories concerning the RSA/RCOD index, the %Dec, and all the others parameters examined (RSA-FT, RSA-TT, RSA-AT, RCOD-FT, RCOD-TT, and RCOD-AT).

The ability to perform RCOD appears to be a more complex movement than RSA. It has been described by Wong et al. (34) and Brughelli et al. (3) that straight sprint and COD performance induced different motor qualities and concluded that there should be similar differences concerning the performance in RSA and RCOD. It could be suggested that RSA and RCOD performance depend on the players' skill level. Indeed, Impellizzeri et al. (18) and Rampinini et al. (24) found better RSA performance in professional players compared with amateurs players, which could differentiate the match performance and the apparition of fatigue according to the different playing level (20). Moreover, the agility of the players appears also to affect the performance in RCOD especially concerning the changes in interlimb coordination, foot control, and ground reaction and contact times, suggesting a specific training of sprint ability, motricity, and adapted strength program (34,35).

However, it could be suggested that the motricity, the agility, the interlimb coordination, and the ability to perform the smallest ground contact is not fully developed for youth soccer players. The results of the present study confirm this by showing a large difference between U15 and U19/Pro2 categories. In this context, coaches have to pay attention on the physiological profiles, chronological age, biological maturity, training age, morphology, anthropometry characteristics, and

playing level to estimate which factor could affect the RSA and RCOD young players performance (17,19,34). The RSA and RCOD depend on neuromuscular and metabolic factors and it was showed that a 10-week program on RSA (2–3 set with 5–6 repetition of 15–20 m interspaced by 14 or 23 seconds of recovery) improved the RSA-FT, RSA-AT, and RCOD in soccer players from a U15 category (4). Therefore, coaches of young soccer players should include specific strength training to improve the RSA and RCOD

performance (6). However, COD and RCOD need complementary training to be consistently improved because the effects of straight sprint and COD were found to be nontransferable (3). A specific program should include neuromuscular solicitation (neural drive, motor unit activation, synchronization inter- and intramuscular), specific strength, intermittent exercises with shuttle (to improve oxidative capacity of enzymes), plyometric training, RCOD with different angle of turn, and maximal concentration to improve the reactivity, the ground reaction time, and the RCOD.

The COD implies running actions in which players must sharply decelerate and block, requiring eccentric muscular efforts and increasing energy cost (6). Additionally, players must reaccelerate and this action solicits essentially the anaerobic metabolism and the fast twitch muscle fibers which possess less mitochondrial and oxidative enzymes than slow twitch but they present more glycolytic enzymes (phosphofructokinase, lactate dehydrogenase, and malate dehydrogenase) (8). Anaerobic glycolysis would take part more strongly and more quickly in the energy metabolism than a straight RSA exercise, and therefore, recovery periods have to be enough long to sufficiently metabolize lactate and restore phosphocreatine. Consequently, high-intermittent exercises appears to be the best mode to improve all the aforementioned metabolic and neuromuscular factors during COD, especially by improving the H^+ buffering, ground contact time, specific strength, and to restore the phosphocreatine.

PRACTICAL APPLICATIONS

This study is the first demonstrating that RSA and RCOD are age dependent in preformation (U15) and formation (U17, U19, and Pro2) in elite soccer. Therefore, coach may use the performance and the RSA/RCOD index to guide the training priority at different age groups. Moreover, taking into account the maturity and biological profiles of the players, coaches should individualize the program between players especially since U17 when the positional roles are clearly determined. Thus, U15 requires a general program

soliciting the neuromuscular and metabolic component to improve simultaneously the RSA and RCOD performance, whereas in U17, U19, and Pro2, the training should be adapted according to the different playing position and the technical-tactical instructions of the coach.

ACKNOWLEDGMENTS

The authors have no conflicts of interest that are directly relevant to the content of this article. This study was not supported by any financial aid. Results of the present study do not constitute endorsement of the product by the authors.

REFERENCES

- Bishop, D, Girard, O, and Mendez-Villanueva, A. Repeated-sprint ability—Part II: Recommendations for training. *Sports Med* 41: 741–756, 2011.
- Bloomfield, J, Polman, R, and O'Donoghue, P. Deceleration movements performed during FA Premier League soccer matches. *J Sports Sci Med* 6(Suppl. 10): 6, 2007.
- Brughelli, M, Cronin, J, Levin, G, and Chaouachi, A. Understanding change of direction ability in sport: A review of resistance training studies. *Sports Med* 38: 1045–1063, 2008.
- Buchheit, M, Mendez-Villanueva, A, Delhomel, G, Brughelli, M, and Ahmaidi, S. Improving repeated sprint ability in young elite soccer players: Repeated shuttle sprints vs. explosive strength training. *J Strength Cond Res* 24: 2715–2722, 2010.
- Dellal, A, Chamari, C, Wong, DP, Ahmaidi, S, Keller, D, Barros, MLR, Bisciotti, GN, and Carling, C. Comparison of physical and technical performance in European professional soccer match-play: The FA Premier League and La LIGA. *Eur J Sport Sci* 11: 51–59, 2011.
- Dellal, A, Keller, D, Carling, C, Chaouachi, A, Wong, PL, and Chamari, K. Physiological effects of directional changes in intermittent exercise in soccer players. *J Strength Cond Res* 24: 3219–3322, 2010.
- Drust, B, Waterhouse, J, Atkinson, G, Edwards, B, and Reilly, T. Circadian rhythms in sports performance—An update. *Chronobiol Int* 22: 21–44, 2005.
- Essen, B, Hagenfeldt, L, and Kaijser, L. Utilization of blood-borne and intramuscular substrates during continuous and intermittent exercise in man. *J Physiol* 265: 489–506, 1977.
- Gabbett, TJ. The development of a test of repeated-sprint ability for elite women's soccer players. *J Strength Cond Res* 24: 1191–1194, 2010.
- Gil, SM, Gil, J, Ruiz, F, Irazusta, A, and Irazusta, J. Physiological and anthropometric characteristics of young soccer players according to their playing position: Relevance for the selection process. *J Strength Cond Res* 21: 438–445, 2007.
- Girard, O, Mendez-Villanueva, A, and Bishop, D. Repeated-sprint ability—Part I. Factors contributing to fatigue. *Sports Med* 41: 673–694, 2011.
- Glaister, M. Multiple sprint work physiological responses, mechanisms of fatigue and the influence of aerobic fitness. *Sports Med* 35: 757–777, 2005.
- Glaister, M. Multiple-sprint work: Methodological, physiological, and experimental issues. *Int J Sports Physiol Perform* 3: 101–112, 2008.
- Glaister, M, Howatson, G, Pattison, JR, and McInnes, G. The reliability and validity of fatigue measures during multiple-sprints work: An issue revisited. *J Strength Cond Res* 22: 1597–1601, 2008.
- Haj-Sassi, R, Dardouri, W, Gharbi, Z, Chaouachi, A, Mansour, H, Rabhi, A, and Mahfoudhi, ME. Reliability and validity of a new repeated agility test as measure of anaerobic and explosive power. *J Strength Cond Res* 25: 472–480, 2011.
- Harriss, DJ and Atkinson, G. International Journal of Sports Medicine—Ethical standards in sport and exercise science research. *Int J Sports Med* 30: 701–702, 2009.
- Hirose, N. Prediction of talent in youth soccer players: Prospective study over 4–6 years. *Football Sci* 8: 1–7, 2011.
- Impellizzeri, FM, Rampinini, E, Castagna, C, Bishop, D, Ferrari Bravo, D, Tibaudi, A, and Wisloff, U. Validity of a repeated sprint test for football. *Int J Sports Med* 29: 899–905, 2008.
- Lago-Penas, C, Casais, L, Dellal, A, Rey, E, and Dominguez, E. Anthropometric and physiological characteristics of young soccer players according to their playing positions: Relevance for competition success. *J Strength Cond Res* 25: 3358–3367, 2011.
- Mohr, M, Krustup, P, and Bangsbo, J. Match performance of high-standard soccer players with special reference to development of fatigue. *J Sports Sci* 21: 519–528, 2003.
- Pyne, DB, Saunders, PU, Montgomery, PG, Hewitt, AJ, and Sheehan, K. Relationship between repeated sprint testing, speed and endurance. *J Strength Cond Res* 22: 1633–1637, 2008.
- Rampinini, E, Bishop, D, Marcora, SM, Ferrari Bravo, D, Sassi, R, and Impellizzeri, FM. Validity of simple field tests as indicators of match-related physical performance in top-level professional soccer players. *Int J Sports Med* 28: 228–235, 2007.
- Rampinini, E, Impellizzeri, FM, Castagna, C, Coutts, AJ, and Wisloff, U. Technical performance during soccer matches of the Italian Serie A league: Effect of fatigue and competitive level. *J Sci Med Sport* 12: 227–233, 2009.
- Rampinini, E, Sassi, A, Morelli, A, Mazzoni, S, Franchini, M, and Coutts, AJ. Repeated-sprint ability in professional and amateur soccer players. *Appl Physiol Nutr Metab* 34: 1048–1054, 2009.
- Reilly, T. An ergonomics model of the soccer training process. *J Sports Sci* 23: 561–572, 2005.
- Reilly, T, Bangsbo, J, and Franks, A. Anthropometric and physiological predispositions for elite soccer. *J Sports Sci* 18: 669–683, 2000.
- Reilly, T, Williams, AM, Nevill, A, and Franks, A. A multidisciplinary approach to talent identification in soccer. *J Sports Sci* 18: 695–702, 2000.
- Salaj, S and Markovic, G. Specificity of jumping, sprinting, and quick change-of-direction motor abilities. *J Strength Cond Res* 25: 1249–1255, 2011.
- Segui, R, Dellal, A, Pialoux, and Hautier, C. The D-test: A new tool for assessing the agility of young soccer players in pre-formation categories. Acta of 17th Annual ECSS-Congress, Bruges, 2012.
- Sheppard, JM and Young, WB. Agility literature review: Classifications, training and testing. *J Sports Sci* 24: 919–932, 2006.
- Spencer, M, Bishop, D, Dawson, B, and Goodman, C. Physiological and metabolic responses of repeated-sprint activities: Specific to field-based team sports. *Sports Med* 35: 1–18, 2005.
- Spencer, M, Lawrence, D, Rechichi, C, Bishop, D, Dawson, B, and Goodman, C. Time-motion analysis of elite field hockey, with special reference to repeated-sprint activity. *J Sport Sci* 22: 843–850, 2004.
- Stølen, T, Chamari, K, Castagna, C, and Wisloff, U. Physiology of soccer: An update. *Sports Med* 35: 501–536, 2005.
- Wong, DP, Chan, GS, and Smith, AW. Repeated sprint and change-of-direction abilities in physically active individuals and soccer players: Training and testing implications. *J Strength Cond Res* 26(9): 2324–2330, 2012.
- Young, WB, McDowell, MH, and Scarlett, BJ. Specificity of sprint and agility training methods. *J Strength Cond Res* 15: 315–319, 2001.